

Администрирование сетевых подсистем

Лабораторная работа №12

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Цели и задачи работы

Получение навыков по управлению системным временем и настройке синхронизации времени с использованием службы Chrony.

Выполнение лабораторной работы

Проверка конфигурации времени

```
[manovichkov@server.manovichkov.net ~]$ timedatectl
    Local time: Sat 2025-10-25 09:59:40 UTC
    Universal time: Sat 2025-10-25 09:59:40 UTC
        RTC time: Sat 2025-10-25 09:59:40
    Time zone: UTC (UTC, +0000)
System clock synchronized: yes
    NTP service: active
    RTC in local TZ: no
[manovichkov@server.manovichkov.net ~]$ date
Sat Oct 25 09:59:44 AM UTC 2025
[manovichkov@server.manovichkov.net ~]$ hwclock
hwclock: Cannot access the Hardware Clock via any known method.
hwclock: Use the --verbose option to see the details of our search for an access method.
```

Рис. 1: Проверка времени на сервере

```
[manovichkov@client.manovichkov.net ~]$ sudo -i
[sudo] password for manovichkov:
[root@client.manovichkov.net ~]# timedatectl
          Local time: Sat 2025-10-25 10:01:52 UTC
        Universal time: Sat 2025-10-25 10:01:52 UTC
           RTC time: Sat 2025-10-25 10:01:52
          Time zone: UTC (UTC, +0000)
System clock synchronized: yes
            NTP service: active
        RTC in local TZ: no
[root@client.manovichkov.net ~]# date
Sat Oct 25 10:01:58 AM UTC 2025
[root@client.manovichkov.net ~]# hwclock
2025-10-25 10:02:01.792079+00:00
[root@client.manovichkov.net ~]# █
```

Рис. 2: Проверка времени на клиенте

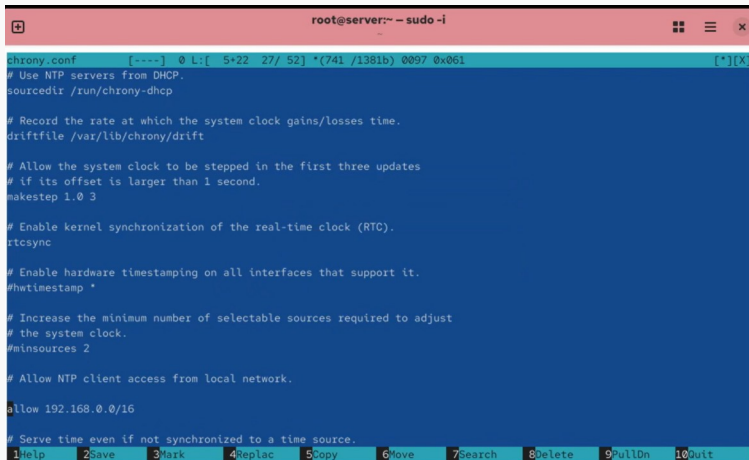
```
[root@server.manovichkov.net ~]# dnf -y install chrony
Extra Packages for Enterprise Linux 10 - x86_64      85 kB/s | 34 kB      00:00
Extra Packages for Enterprise Linux 10 - x86_64      6.0 MB/s | 4.8 MB      00:00
Rocky Linux 10 - BaseOS                             387 B/s | 4.3 kB      00:11
Rocky Linux 10 - BaseOS                             3.7 MB/s | 21 MB      00:05
Rocky Linux 10 - AppStream                          14 kB/s | 4.3 kB      00:00
Rocky Linux 10 - AppStream                          7.1 MB/s | 2.2 MB      00:00
Rocky Linux 10 - CRB                                3.4 kB/s | 4.3 kB      00:01
Rocky Linux 10 - CRB                                1.2 MB/s | 527 kB      00:00
Rocky Linux 10 - Extras                             12 kB/s | 3.1 kB      00:00
Rocky Linux 10 - Extras                             21 kB/s | 5.4 kB      00:00
Package chrony-4.6.1-1.el10.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@server.manovichkov.net ~]# █
```

Рис. 3: Источники времени на сервере

```
[root@server.manovichkov.net ~]# chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^- 195.122.253.38             2    6   377    36  +3755us[+3755us] +/-   30ms
^* mail.iotserv.ru           2    6   377    43   +761us[+1478us] +/-  5642us
^- 91.235.144.128            1    6   377    51   -133us[ +585us] +/-   55ms
^- 188.225.33.239            2    6   377    50   -490us[ -490us] +/-   17ms
[root@server.manovichkov.net ~]#
```

Рис. 4: Источники времени на клиенте

Настройка сервера синхронизации



```
root@server:~ - sudo -i
chrony.conf      [----] 0 L:[ 5+22 27/ 52] *(741 /1381b) 0097 0x061 [*][X]
# Use NTP servers from DHCP.
sourcedir /run/chrony-dhcp

# Record the rate at which the system clock gains/losses time.
driftfile /var/lib/chrony/drift

# Allow the system clock to be stepped in the first three updates
# if its offset is larger than 1 second.
makestep 1.0 3

# Enable kernel synchronization of the real-time clock (RTC).
rtcsync

# Enable hardware timestamping on all interfaces that support it.
#hwtimestamp *

# Increase the minimum number of selectable sources required to adjust
# the system clock.
#minsources 2

# Allow NTP client access from local network.
allow 192.168.0.0/16

# Serve time even if not synchronized to a time source.
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

```
[root@server.manovichkov.net ~]# systemctl restart chronyd
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# firewall-cmd --add-service=ntp --per
success
[root@server.manovichkov.net ~]# firewall-cmd --reload
success
[root@server.manovichkov.net ~]#
```

Рис. 6: Перезапуск службы chronyd и настройка firewall

Настройка клиента синхронизации

```
root@client:~ - sudo -i
chrony.conf  [----] 36 L:[ 1+ 4 5/ 53] *(196 /1420b) 0010 0x00A [*][X]
# Use public servers from the pool.ntp.org project.
# Please consider joining the pool (https://www.pool.ntp.org/join.html).
#pool 2.rocky.pool.ntp.org iburst

server server.manovichkov.net iburst

# Use NTP servers from DHCP.
#sourceaddr /run/chrony-dhcp

# Record the rate at which the system clock gains/losses time.
#driftfile /var/lib/chrony/drift

# Allow the system clock to be stepped in the first three updates
# if its offset is larger than 1 second.
#makestep 1.0 3

# Enable kernel synchronization of the real-time clock (RTC).
#rtcsync

# Enable hardware timestamping on all interfaces that support it.
#hwtimestamp *

# Increase the minimum number of selectable sources required to adjust
# the system clock.

1help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn10Quit
```

```
[root@client.manovichkov.net ~]# systemctl restart chronyd
[root@client.manovichkov.net ~]#
[root@client.manovichkov.net ~]# chronyc sources
MS Name/IP address          Stratum Poll Reach LastRx Last sample
=====
^? dhcp.manovichkov.net      3      6      3      0  -2639us[-2639us] +/-  16ms
[root@client.manovichkov.net ~]#
```

Рис. 8: Результат синхронизации клиента

Выводы по проделанной работе

В ходе лабораторной работы были изучены и настроены механизмы синхронизации времени в операционной системе Rocky Linux.

Установлен и сконфигурирован пакет `chrony`, обеспечивающий работу службы NTP. Проверена корректность системного и аппаратного времени, выполнена настройка локального сервера синхронизации и клиента.